

Part A: Directory and File Management

Directory Structure:

```
STUDENT_ID_assignment:
lesson1

STUDENT_ID_assignment/lesson1:
partA partB partC

STUDENT_ID_assignment/lesson1/partA:
dir1 dir2

STUDENT_ID_assignment/lesson1/partA/dir1:
file1.txt file2.txt file3.txt file4.txt file5.txt subdir1 subdir2

STUDENT_ID_assignment/lesson1/partA/dir1/subdir1:

STUDENT_ID_assignment/lesson1/partA/dir1/subdir2:

STUDENT_ID_assignment/lesson1/partA/dir2:
file6.txt file7.txt file8.txt subdir3 subdir4

STUDENT_ID_assignment/lesson1/partA/dir2/subdir3:

STUDENT_ID_assignment/lesson1/partA/dir2/subdir4:

STUDENT_ID_assignment/lesson1/partB:
subdir5

STUDENT_ID_assignment/lesson1/partB/subdir5:
file10.txt file9.txt

STUDENT_ID_assignment/lesson1/partC:
subdir6

STUDENT_ID_assignment/lesson1/partC/subdir6:
file11.txt
```

Command History:

```
STUDENT_ID_assignment/lesson1/partC/subdir6:
file11.txt
ubuntu@ip-172-31-20-99:~$ history
  1  ls
  2  ls -a
  3  ls -la
  4  ls -lah
  5  clear
  6  ls -a
  7  ls [.cache]
  8  ls .ssh
  9  ls .cache
 10  man ls
 11  clear
 12  ls
 13  mkdir binf5002
 14  ls
 15  ls binf5002
 16  mkdir binf5002/folder1
 17  ls
 18  ls binf5002
 19  mkdir binf5002/folder2
 20  ls binf5002
 21  mkdir binf5002/folder1/child1
 22  ls binf5002/folder1
 23  ls folder1
 24  ls -R
 25  ls -Ra
 26  ls -aR
 27  clear
 28  ls
 29  touch file1.txt
 30  ls binf5002
 31  ls
 32  touch binf5002/file2.txt
 33  ls -R
 34  touch binf5002/folder1/child1/file3.txt
 35  ls -R
 36  clear
 37  ls
 38  cp file1.txt copy.txt
 39  ls
 40  ls binf5002/
 41  cp file1.txt binf5002/copy2.txt
 42  ls binf5002
 43  cp binf5002/folder1/child1/file3.txt binf5002/folder2/copy3.txt
 44  ls -R
```

```
45 cp binf5002 copy_folder
46 cp -r binf5002 copy_folder
47 ls -R
48 clear
49 ls
50 rm file1.txt
51 ls
52 rm copy_folder
53 rm -r copy_folder
54 ls
55 clear
56 ls
57 mv copy.txt binf5002
58 ls
59 ls binf5002
60 mv new.txt rename.txt
61 touch new.txt
62 mv new.txt rename.txt
63 ls
64 mv binf5002/copy.txt moved_copy.txt
65 ls
66 la
67 mkdir STUDENT_ID_assignment/lesson1/partA/dir1/
68 mkdir STUDENT_ID_assignment
69 la
70 mkdir STUDENT_ID_assignment/lesson1
71 mkdir STUDENT_ID_assignment/lesson1/partA
72 mkdir STUDENT_ID_assignment/lesson1/partA/dir1
73 mkdir STUDENT_ID_assignment/lesson1/partA/dir2
74 mkdir STUDENT_ID_assignment/lesson1/partB
75 mkdir STUDENT_ID_assignment/lesson1/partC
76 la STUDENT_ID_assignment
77 la STUDENT_ID_assignment/lesson1
78 mkdir STUDENT_ID_assignment/lesson1/partA/dir1/subdir1
79 mkdir STUDENT_ID_assignment/lesson1/partA/dir1/subdir2
80 touch file1.txt STUDENT_ID_assignments/lesson1/partA/dir1
81 touch file1.txt STUDENT_ID_assignment/lesson1/partA/dir1
82 touch file2.txt STUDENT_ID_assignment/lesson1/partA/dir1
83 touch file3.txt STUDENT_ID_assignment/lesson1/partA/dir1
84 touch file4.txt STUDENT_ID_assignment/lesson1/partA/dir1
85 touch file5.txt STUDENT_ID_assignment/lesson1/partA/dir1
86 mkdir STUDENT_ID_assignment/lesson1/partA/dir2/subdir3
87 mkdir STUDNET_ID_assignment/lesson1/partA/dir2/subdir4
88 mkdir STUDENT_ID_assignment/lesson1/partA/dir2/subdir4
89 touch file6.txt STUDENT_ID_assignment/lesson1/partA/dir2
90 touch file7.txt STUDENT_ID_assignment/lesson1/partA/dir2
91 touch file8.txt STUDENT_ID_assignmnet/lesson1/partA/dir2
```

```
92 touch file8.txt STUDENT_ID_assignment/lesson1/partA/dir2
93 mkdir STUDENT_ID_assignment/lesson1/partB/subdir5
94 touch file9.txt STUDENT_ID_assignment/lesson1/partB/subdir5
95 touch file10.txt STUDENT_ID_assignment/lesson1/partB/subdir5
96 mkdir STUDENT_ID_assignment/lesson1/partC/subdir6
97 touch file11.txt STUDENT_ID_assignment/lesson1/partC/subdir6
98 la -R STUDENT_ID_assignment
99 la -R STUDENT_ID_assignment/lesson1/partC/subdir6
100 la
101 rm file1.txt
102 rm file2.txt
103 rm file3.txt
104 rm file4.txt
105 rm file5.txt
106 rm file6.txt
107 rm file7.txt
108 rm file8.txt
109 rm file9.txt
110 rm file10.txt
111 rm file11.txt
112 la
113 touch STUDENT_ID_assignment/lesson1/partA/dir1/file1.txt
114 touch STUDENT_ID_assignment/lesson1/partA/dir1/file2.txt
115 touch STUDENT_ID_assignment/lesson1/partA/dir1/file3.txt
116 touch STUDENT_ID_assignment/lesson1/partA/dir1/file4.txt
117 touch STUDENT_ID_assignment/lesson1/partA/dir1/file5.txt
118 touch STUDENT_ID_assignment/lesson1/partA/dir2/file6.txt
119 touch STUDENT_ID_assignment/lesson1/partA/dir2/file7.txt
120 touch STUDENT_ID_assignment/lesson1/partA/dir2/file8.txt
121 touch STUDENT_ID_assignment/lesson1/partB/subdir5/file9.txt
122 touch STUDENT_ID_assignment/lesson1/partB/subdir5/file10.txt
123 touch STUDENT_ID_assignment/lesson1/partC/subdir6/file11.txt
124 la -R STUDENT_ID_assignment
125 history
```

Part B: Path Navigation and File Operations

Directory Structure:

```
STUDENT_ID_assignment:
lesson1

STUDENT_ID_assignment/lesson1:
partA partB partC

STUDENT_ID_assignment/lesson1/partA:
dir1 dir2

STUDENT_ID_assignment/lesson1/partA/dir1:
file1.txt file2.txt file3.txt file9.txt subdir1 subdir2

STUDENT_ID_assignment/lesson1/partA/dir1/subdir1:

STUDENT_ID_assignment/lesson1/partA/dir1/subdir2:

STUDENT_ID_assignment/lesson1/partA/dir2:
file6.txt file7_renamed.txt file8.txt subdir3 subdir4

STUDENT_ID_assignment/lesson1/partA/dir2/subdir3:
file1.txt

STUDENT_ID_assignment/lesson1/partA/dir2/subdir4:

STUDENT_ID_assignment/lesson1/partB:
subdir5

STUDENT_ID_assignment/lesson1/partB/subdir5:
file4.txt file9.txt

STUDENT_ID_assignment/lesson1/partC:
subdir6

STUDENT_ID_assignment/lesson1/partC/subdir6:
file11.txt
```

Command History:

```
127 cd STUDENT_ID_assignment/lesson1/partA/dir1/subdir1
128 pwd
129 cd ..
130 la
131 mv file1.txt /partB/subdir5/
132 cd
133 cd -
134* a
135 clear
136 mv file4.txt /STUDENT_ID_assignment/lesson1/partB/subdir5
137 cd
138 mv STUDENT_ID_assignment/lesson1/partA/dir1/file4.txt STUDENT_ID_assignment/lesson1/partB/subdir5
139 la STUDENT_ID_assignment/lesson1/partB/subdir5
140 cd STUDENT_ID_assignment/lesson1/partA/dir2
141 la
142 mv file7.txt file7_renamed.txt
143 la
144 cd ..
145 cd dir1
146 cp file1.txt STUDENT_ID_assignment/lesson1/partA/dir2/subdir3
147 ccp file1.txt /dir2/subdir3
148 cp file1.txt /dir2/subdir3
149 cd
150 cp STUDENT_ID_assignment/lesson1/partA/dir1/file1.txt STUDENT_ID_assignment/lesson1/partA/dir2/subdir3
151 cp STUDENT_ID_assignment/lesson1/partB/subdir5/file9.txt STUDENT_ID_assignment/lesson1/partA/dir1
152 rm STUDENT_ID_assignment/lesson1/partA/dir1/file5.txt
153 rm STUDENT_ID_assignment/lesson1/partB/subdir5/file10.txt
154 la -R STUDENT_ID_assignment
155 history
```

Part C: Documentation

Part 1: Use man hier to read the manual for the file system hierarchy. Explain the purpose of the /mnt, /proc, /sys, /srv, /root, /run folders.

/mnt - This directory is a mount point for a temporarily mounted filesystem. In some distributions, /mnt contains subdirectories intended to be used as mount points for several temporary filesystems.

/proc - This is a mount point for the proc filesystem, which provides information about running processes and the kernel. This pseudo-filesystem is described in more detail in proc(5).

/sys - This is a mount point for the sysfs filesystem, which provides information about the kernel like /proc, but better structured, following the formalism of kobject infrastructure.

/srv - This directory contains site-specific data that is served by this system.

/root - This directory is usually the home directory for the root user (optional).

/run - This directory contains information which describes the system since it was booted. Once this purpose was served by /var/run and programs may continue to use it.

Part 2: Use man to read the manual for the rm command. Summarize purpose and usage of -r and -i options.

-i - prompt before every removal
-r - remove directories and their contexts recursively

Part 3: Use --help for the cp command and note down what each of the following options does: -r, -i, -u.

-r - copy directories recursively
-i - prompt before overwrite (overrides a previous -n option)
-u - equivalent to --update[=older]. See below

Part 4: Command History

```
157 man hier
158 man rm
159 --help
160 --help cp
161 cp --help
162 history
```